機械力学 I 解答用紙

week-06 非減衰自由振動- 重力を考量した場合の運動，単振り子の振動

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m08a06B0101

(1)	9.90×10^1 [rad/s]
(2)	9.80×10^3 [N/m]

m08a06B0201

$\delta = \frac{mg}{k}$ 196 [mm]

m08b03B0102

(1)	$\omega_n = \sqrt{\frac{g}{l}} \fallingdotseq 4.43$ [rad/s]
(2)	$f_n = \frac{\omega_n}{2\pi} = 0.705$ [Hz]

m08b03B0201

$\omega_n = \cancel{\sqrt{\frac{mgh}{I+mh^2}} \fallingdotseq 2.57 \text{ [rad/s]}}$

$\sqrt{\frac{mgh}{I}} \fallingdotseq 2.71$ [rad/s]

m08b04B0201

(1)	$T_g = -mgh\sin\theta$
(2)	$T = T_g = -mgh\sin\theta$
(3)	$I = I_d + mh^2$
(4)	$I\ddot{\theta} + mgh\sin\theta = 0$
(5)	$I\ddot{\theta} + mgh\theta = 0$
(6)	$-I\omega_n^2 + mgh = 0$
(7)	$\omega_n = \sqrt{\frac{mgh}{I_d + mh^2}}$

フリースペース（質問などどうぞ）